



Universität Osnabrück
Master's Program in Business Informatics

**Design, Implementation, and Evaluation of a
Simulation-Based Benchmark for Comparing
Approaches to Robot Navigation**

A Design Science Research Approach to Capabilities, Limitations, and Comparative
Evaluation

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Exposé for the Master's Thesis

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1 Introduction

Robots are increasingly expected to navigate through structured, semi-structured, and changing environments using sensor data, simulation environments, and artificial intelligence. In practical navigation scenarios, system quality cannot be assessed reliably through one isolated success criterion. A robot may reach a target position while drifting in localization, following an inefficient trajectory, ignoring semantic goal information, failing under changed environmental conditions, or reacting poorly to unexpected obstacles. The central challenge is therefore not only movement execution, but **robust navigation under defined and reproducible evaluation conditions**.

This exposé proposes a master’s thesis on **simulation-based benchmarking of approaches to robot navigation** (*Simulationsbasiertes Benchmarking von Ansätzen zur Roboternavigation*). The work will compare major approach classes that are relevant for navigation, such as **Simultaneous Localization and Mapping (SLAM)**, **Vision-Language-Action models (VLA)**, **Reinforcement Learning (RL)**, and **World Models (WM)**. These approaches represent different assumptions about what is most important for navigation. SLAM focuses on spatial localization, mapping, and pose consistency. VLA models connect visual observations and language or goal descriptions to possible robot actions. RL focuses on learning navigation or control behavior through reward-based interaction. World Models learn predictive representations of environment dynamics and can support planning, robustness analysis, or early recognition of navigation-relevant deviations.

A direct comparison is scientifically interesting but must be framed carefully. SLAM, VLA, RL, and World Models are not interchangeable algorithms for one identical subtask. They solve partially overlapping but conceptually different problems. SLAM is primarily a spatial perception and state-estimation method, VLA is primarily an instruction-conditioned action-generation paradigm, RL is primarily a learning paradigm for policy optimization, and WM is primarily a predictive dynamics-modeling paradigm. The thesis therefore does not aim to declare one universally superior method. Instead, it investigates **which approach is suitable for which navigation capability, under which limitations, and with which practical trade-offs**.

The planned thesis follows a **Design Science Research** methodology. This is appropriate because the work creates and evaluates an artifact rather than only describing existing systems. The central artifact is a simulation-based benchmark demonstrator for robot navigation. The artifact will include a still-to-be-finalized navigation scenario, a data-generation and logging pipeline, perturbation mechanisms, model-training or model-adaptation pipelines for the relevant approach classes, and an evaluation framework. Following the Design Science Research logic, the artifact is designed, implemented, demonstrated, and evaluated systematically (Hevner et al. 2004; Peffers et al. 2007).

The exact use case is not defined at this stage. This is intentional because the thesis should not be restricted too early to a specific latent-space formulation or predetermined application scenario. Possible use cases include navigation in a warehouse-like layout, corridor or room navigation, mobile robot navigation with static and dynamic obstacles, navigation toward language-described target zones, or a simplified navigation-support scenario in which navigation is the central challenge. The final use case should be selected according to technical feasibility, available simulation assets, robot model support, and its ability to expose meaningful differences between the compared approaches.

2 Objective

The objective of this master’s thesis is to design, implement, train or adapt, and evaluate a simulation-based benchmark demonstrator that enables a structured comparison of approaches to robot navigation. The demonstrator serves as the central Design Science artifact. A central responsibility within the thesis is the training or adaptation of the learning-based models used in the benchmark, especially RL policies, World Model components, and, where feasible, VLA-based or VLA-inspired modules. For non-learning components such as classical SLAM, the corresponding work concerns configuration, calibration, integration, and benchmark evaluation rather than neural model training.

The scientific objective is to clarify the relative strengths and weaknesses of the selected approach classes in a controlled simulation-based navigation scenario. The comparison focuses on localization and mapping capability, goal or instruction grounding, learned control performance, trajectory quality, predictive dynamics, robustness under perturbations, data efficiency, implementation effort, interpretability, reproducibility, and transferability to real robotic settings.

The practical objective is to create a reproducible evaluation setup in which navigation episodes can be generated, logged, used for model training, analyzed, and compared. The demonstrator should make it possible to answer questions such as: Does SLAM provide strong localization and mapping but limited semantic goal understanding? Do VLA models generalize better to language-conditioned or goal-conditioned changes but introduce reliability and deployment challenges? Does RL produce robust navigation behavior after training but require many simulation interactions? Do World Models provide useful predictive signals for planning, deviation recognition, or early-warning behavior? How much training data, compute effort, and engineering effort is required to make each learning-based approach usable in the demonstrator?

The expected result is not a single ranking of methods, but a differentiated evaluation matrix. The thesis should show which navigation capabilities each approach supports, where its limitations become visible, and whether hybrid designs are more promising than isolated methods.

3 State of Research

3.1 Robot Navigation, Benchmarking, and Deviation Detection

Robot navigation research covers localization, mapping, path planning, control, semantic grounding, and robustness under environmental variation. Benchmarking is essential because different navigation approaches often optimize different goals and are evaluated with different metrics. Classical monitoring and deviation-detection methods such as rule-based checks, threshold-based monitoring, statistical process control, and visual deviation detection remain relevant because they are interpretable and comparatively easy to implement (Bergmann et al. 2019; Montgomery 2019).

However, navigation failures are frequently spatial, temporal, behavioral, and semantic at the same time. A robot can fail because it loses localization, builds an inconsistent map, chooses an inefficient path, collides with obstacles, reacts poorly to dynamic changes, or does not act according to a goal instruction. These cases require evaluation methods that go beyond a single

success-rate metric and instead compare capabilities, limitations, and practical implementation requirements.

3.2 SLAM and Spatial Navigation Monitoring

SLAM addresses the problem of estimating a robot’s pose while simultaneously building or maintaining a map of the environment (Durrant-Whyte and Bailey 2006). Modern systems such as ORB-SLAM3 support visual, visual-inertial, and multi-map SLAM with monocular, stereo, and RGB-D cameras (Campos et al. 2021). In robot navigation, SLAM-related methods are relevant because they provide spatial consistency, robot localization, map stability, and trajectory information.

Depending on the final use case, full SLAM may not always be necessary. For a simple simulated environment, visual odometry, pose tracking, or ground-truth-assisted localization may be sufficient as a baseline. Nevertheless, SLAM remains an important comparison category because robot navigation often depends on localization, mapping, spatial consistency, and robustness in changing workspaces or spatially extended environments.

In this thesis, SLAM is treated as a **spatial navigation baseline**. Its expected strength is localization, mapping, trajectory consistency, and the detection of geometric deviations. Its expected weakness is that it does not directly understand semantic goals, high-level instructions, adaptive decision-making, or long-horizon navigation success without additional logic.

3.3 Vision-Language-Action Models

Vision-Language-Action models connect visual observations, language instructions, and robot actions. Recent work such as RT-2 investigates how vision-language models trained with large-scale web data can be incorporated into end-to-end robotic control (Zitkovich et al. 2023). Open X-Embodiment and RT-X demonstrate the relevance of large cross-robot datasets for generalist robot policies (Open X-Embodiment Collaboration et al. 2024). OpenVLA provides an open-source VLA model trained on a large collection of robot demonstrations and explicitly targets generalizable visuomotor control (Kim et al. 2025). Octo similarly contributes an open-source generalist robot policy that can be adapted to different robot setups (Octo Model Team et al. 2024).

For the proposed thesis, VLA models are relevant because robots are increasingly expected to react flexibly to high-level instructions, goal descriptions, landmarks, and changing task contexts. A VLA approach can potentially evaluate whether the current scene and action match a navigation instruction such as “move to the target zone while avoiding the obstacle near the left shelf”. This gives VLA models a semantic advantage over purely geometric approaches.

However, VLA methods also introduce practical limitations. They may require large pretrained models, careful action-space adaptation, GPU resources, and robust grounding between language, perception, and robot control. In this thesis, the training responsibility for VLA does not necessarily mean pretraining a foundation model from scratch. Instead, the feasible scope may include fine-tuning an open VLA model, adapting an existing model to the simulated robot action space, training a smaller VLA-inspired module, or combining visual-language interpretation with a classical controller. The evaluation will explicitly document the chosen operationalization

and distinguish between pretrained knowledge, thesis-specific adaptation, and newly trained components.

3.4 Reinforcement Learning for Robot Navigation and Control

Reinforcement Learning studies how agents learn behavior through interaction with an environment and reward feedback (Sutton and Barto 2018). Deep RL has shown that policies can be learned from high-dimensional observations (Mnih et al. 2015). Policy-gradient and actor-critic methods such as PPO are widely used in simulated robot learning (Schulman et al. 2017). Simulation environments such as Isaac Lab are particularly relevant because they allow many training episodes and systematic perturbation of tasks.

In this thesis, RL is treated as a **learned navigation and control paradigm**. The RL policy will be trained within the simulation environment using a defined reward function, action space, observation space, and training protocol. Its expected strength is that it can learn robust navigation behavior from interaction, especially when the task can be specified through rewards. Its expected weakness is sample complexity, sensitivity to reward design, limited interpretability, and possible poor generalization outside the training distribution. RL may perform well at executing a task but may not automatically explain why a navigation attempt failed unless explicit diagnostic signals are designed.

3.5 World Models and Predictive Dynamics

World Models learn internal representations of an environment and its dynamics. Early work showed how agents can use compressed spatial and temporal representations for behavior learning (Ha and Schmidhuber 2018). DreamerV3 demonstrates the relevance of learned world models for general model-based reinforcement learning across diverse domains (Hafner et al. 2023). Joint-Embedding Predictive Architectures such as V-JEPA learn by predicting representations instead of reconstructing raw pixels, which makes them relevant for video understanding and physical prediction (Bardes et al. 2024). V-JEPA 2 further connects self-supervised video learning with prediction, planning, and robot control (Assran, Bardes, et al. 2025).

For robot navigation, the central idea is that a World Model can learn how environment states, robot states, and action consequences are expected to evolve. If predicted developments differ substantially from observed developments, the discrepancy can be used as a signal for planning uncertainty, deviation recognition, or early-warning behavior. This makes World Models especially relevant for navigation problems that require temporal prediction rather than purely reactive control.

In this thesis, World Models are treated as a **predictive dynamics and planning-support paradigm**. The World Model will be trained on simulation-generated navigation sequences, preferably with a focus on nominal behavior and controlled perturbations for validation and testing. Its expected strength is modeling expected state transitions, anticipating future navigation states, and identifying deviations from expected dynamics. Its expected weakness is training complexity, reduced interpretability compared with explicit rules, and possible sensitivity to simulation realism and representation quality.

3.6 Existing Dataset and Benchmark Candidates

A central design question is whether the comparison can rely on an existing common dataset. Current robotics datasets and benchmarks provide important partial coverage, but they do not fully cover the requirements of this thesis. Open X-Embodiment aggregates large-scale robot demonstration data across multiple robot embodiments and skills and is therefore highly relevant for VLA-style pretraining and adaptation (Open X-Embodiment Collaboration et al. 2024). DROID provides diverse real-world manipulation demonstrations across many scenes and tasks, which makes it useful as an external reference for data diversity and real-world generalization (Khazatsky et al. 2024). RLBench provides a simulated manipulation benchmark with 100 tasks, visual/proprioceptive observations, and automatically generated demonstrations, making it relevant for imitation learning, RL, and task-level benchmarking (James et al. 2020). CALVIN focuses on long-horizon language-conditioned manipulation and is relevant for VLA-style and language-conditioned policy evaluation (Mees et al. 2022). RoboCasa and RoboCasa365 provide large-scale simulation environments, tasks, and demonstration datasets for generalist robot learning, especially in household manipulation scenarios (Soroush Nasiriany, Maddukuri, et al. 2024; Soroush Nasiriany, Sepehr Nasiriany, et al. 2026).

These resources are valuable, but none of them directly provides all modalities required for a fair comparison of SLAM, VLA, RL, and World Models in a navigation-oriented benchmark with controllable perturbations. A suitable common benchmark for this thesis requires synchronized RGB or RGB-D data, camera poses, robot states, actions, language or symbolic goals, navigation-state labels, perturbation or failure labels, event onset times, and access to an interactive environment for RL training and evaluation. Therefore, the thesis will use a **project-specific common dataset generated in simulation** as the primary benchmark. Public datasets may still be used as secondary references, for example for VLA pretraining/fine-tuning, benchmark design, data-format inspiration, or external validity discussion, but the main comparison will be based on one common simulation-generated episode schema.

3.7 Research Gap

Existing research often evaluates SLAM, VLA, RL, and World Models separately. SLAM is commonly evaluated through localization and mapping accuracy. VLA models are often evaluated through navigation success and generalization to instructions, goals, or landmarks. RL is evaluated through accumulated reward, navigation success, sample efficiency, and robustness. World Models are evaluated through prediction quality, planning performance, or representation learning metrics. What is missing is a structured comparison of these paradigms in one simulation-based navigation benchmark under a shared evaluation framework that also documents how the learning-based components are trained, adapted, validated, and compared under similar data and compute constraints.

This thesis addresses this gap by designing a simulation-based demonstrator that operationalizes the four paradigms in a comparable setting. The contribution is not merely technical implementation, but the design of a fair comparison framework that makes clear which navigation capabilities each method provides, which limitations remain, and which practical constraints affect implementation.

4 Research Concept

4.1 Research Questions

The thesis is guided by the following two research questions:

RQ1: How can a simulation-based Design Science Research artifact be designed, implemented, and evaluated to benchmark approaches to robot navigation under reproducible conditions?

RQ2: Which capabilities, limitations, and practical trade-offs do SLAM, VLA, RL, and World Model approaches show in a shared robot-navigation benchmark?

4.2 Hypotheses

The empirical evaluation will be guided by a reduced set of working hypotheses. They are intentionally formulated at the capability level because the compared approaches solve different sub-problems of robot navigation.

- H1:** Classical spatial approaches, especially SLAM-based components, are expected to provide strong localization, mapping, pose consistency, and trajectory assessment, but they are expected to require additional modules for semantic goal interpretation and adaptive decision-making.
- H2:** Learning-based approaches, especially RL, VLA-inspired modules, and World Models, are expected to improve adaptive, semantic, or predictive navigation capabilities, but only at the cost of higher data demand, training effort, compute usage, and implementation complexity.
- H3:** A capability-based or hybrid benchmark design is expected to provide a more meaningful evaluation of robot-navigation approaches than a single aggregated ranking, because each approach contributes different strengths and limitations to the navigation pipeline.

4.3 Methodology: Design Science Research

This thesis follows a **Design Science Research** methodology. Design Science Research is suitable because the thesis creates and evaluates an artifact that addresses a relevant practical and scientific problem. According to the Design Science paradigm, knowledge is generated through the design, construction, application, and evaluation of an artifact (Hevner et al. 2004). The process model by Peffers et al. (2007) structures Design Science Research into problem identification, objective definition, design and development, demonstration, evaluation, and communication.

The artifact in this thesis is a **simulation-based comparative benchmark demonstrator**. It consists of the following components:

- a simulated robot navigation scenario that is still deliberately left flexible;
- a data-generation and logging pipeline;

- nominal navigation episodes and controlled perturbation scenarios;
- operationalized comparison modules for SLAM, VLA, RL, and WM approaches;
- training, fine-tuning, or adaptation pipelines for the learning-based modules;
- evaluation scripts and metrics;
- a structured analysis of quantitative and qualitative results.

The Design Science Research process is applied as follows:

1. **Problem identification and motivation:** Robot navigation requires spatial, semantic, control-related, and temporal capabilities. Existing approaches usually address only parts of this problem.
2. **Definition of objectives:** The objective is to design an artifact that enables a fair and reproducible comparison of SLAM, VLA, RL, and WM approaches in a controlled simulation scenario.
3. **Design and development:** The demonstrator is designed and implemented. This includes simulation setup, navigation scenario definition, sensor configuration, perturbation generation, data logging, method implementation, model training or adaptation, and evaluation tooling.
4. **Demonstration:** The artifact is demonstrated on a representative simulated navigation scenario. The exact use case remains open at this stage and will be selected according to feasibility, available robot models, and the capability dimensions that should be evaluated.
5. **Evaluation:** The implemented and trained approaches are evaluated against shared criteria. The evaluation includes navigation success, localization or pose accuracy, trajectory quality, semantic goal fulfillment, robustness, predictive or deviation-recognition performance, training data requirements, compute effort, and implementation complexity.
6. **Communication:** The artifact design, experimental results, methodological decisions, limitations, and implications are documented in the final master’s thesis.

4.4 Simulation Use Case

The exact use case is not defined at this stage. The thesis is therefore framed as **simulation-based benchmarking of approaches to robot navigation** rather than as a study of one fixed application scenario. The selected scenario should be simple enough to implement reliably within a master’s thesis but rich enough to expose differences between SLAM, VLA, RL, and WM approaches.

Possible scenarios include a mobile robot navigating through a warehouse-like layout, corridor or room navigation with static and dynamic obstacles, navigation toward language-described target zones, or a simplified support task in which navigation is the central challenge. The final choice should be made according to technical feasibility, available simulation assets, robot model support, and the ability to generate reproducible evaluation episodes.

The scenario can be varied through map layout, obstacle positions, lighting, camera viewpoint, target location, goal instruction, sensor noise, and dynamic changes in the environment. Nominal

episodes represent successful navigation under defined conditions. Perturbed or challenging episodes can include:

- localization drift or pose-estimation errors;
- map inconsistency or environmental layout changes;
- obstacle appearance, disappearance, or displacement;
- inefficient, unsafe, or unstable trajectories;
- failure to reach the target zone;
- collision or near-collision events;
- semantic goal or instruction mismatch;
- abnormal timing, hesitation, or recovery behavior;
- physically implausible or unexpected state changes;
- reduced robustness under changed lighting, sensor noise, or unseen layouts.

4.5 Approaches to Be Compared

The comparison will use four primary method classes. The exact implementation can be adjusted during the thesis according to feasibility, but the comparison logic remains fixed. Because model training and adaptation are part of the research artifact, the benchmark will explicitly separate *model training*, *model adaptation*, and *model evaluation*. This is important because SLAM mainly requires configuration and calibration, whereas VLA, RL, and WM components require different degrees of training or fine-tuning.

Approach	Core capability	Possible operationalization	Expected evaluation focus
SLAM	Spatial localization, mapping, pose consistency	ORB-SLAM3 or visual odometry / pose tracking configured with simulated camera data and ground truth	Trajectory error, map consistency, localization robustness, spatial deviation detection
VLA	Vision-language-conditioned action selection and semantic grounding	OpenVLA/Octo-style model, fine-tuned policy, trained smaller VLA-inspired model, or modular VLM plus controller approximation	Navigation success under language variation, semantic correctness, instruction mismatch detection, landmark or target generalization, adaptation effort

Approach	Core capability	Possible operationalization	Expected evaluation focus
RL	Reward-based policy learning and robust control	PPO/SAC-style policy trained in Isaac Lab, MuJoCo/MJX, or a comparable simulation environment	Navigation success, reward, robustness to perturbations, sample efficiency, recovery behavior, training stability
WM	Predictive dynamics and planning support	Dreamer-style dynamics model, JEPA-inspired predictive model, or custom encoder-transition-predictor architecture trained on simulated navigation sequences	Prediction error, deviation recognition, temporal consistency, planning or early-warning ability, training data demand

Table 1: Planned comparison structure for SLAM, VLA, RL, and WM approaches.

4.6 Model Training Scope

The thesis must define a realistic and transparent training scope for all trainable or adaptable components. The term “training” is used differently across the four approach classes. For RL, it refers to training a policy through interaction with the simulated environment. For World Models, it refers to training an encoder and predictive dynamics model on generated navigation sequences. For VLA, it may refer to fine-tuning, action-space adaptation, or training a smaller VLA-inspired module, because pretraining a large-scale foundation model from scratch is outside the expected scope of a master’s thesis. For SLAM, there is normally no comparable neural training phase; instead, the work concerns system configuration, calibration, parameter tuning, and integration into the evaluation pipeline.

The training pipeline will include dataset generation, preprocessing, train-validation-test splits, hyperparameter documentation, checkpoint management, evaluation scripts, and reproducibility documentation. For each trainable component, the thesis will report the training data used, the training objective, the compute resources, the number of training steps or episodes, and the resulting model performance. This makes the comparison more transparent because performance will be interpreted in relation to the effort required to obtain the trained model.

4.7 Data

The primary data source will be simulation-generated navigation episodes. Depending on the final implementation, the dataset may include:

- RGB image sequences;
- depth images or point clouds;

- camera poses and robot poses;
- obstacle, landmark, and target positions;
- robot states, control commands, and trajectories;
- language instructions, goal descriptions, or symbolic navigation tasks;
- action logs;
- navigation-state or progress labels;
- perturbation, deviation, or failure labels and onset times;
- ground-truth success or failure labels.

The dataset will be split into training, validation, and test scenarios. Nominal data will be used to train models that learn expected navigation behavior. Perturbed or challenging episodes will be used for validation and evaluation, depending on the method and task. Scenario randomization should be included to prevent models from only memorizing one fixed layout. For RL, the data concept is complemented by interaction episodes generated during training; for VLA and WM components, stored simulation sequences and annotations form the main training basis.

The primary dataset will be organized as a **common episode dataset**. Each episode represents one complete or interrupted navigation attempt and contains synchronized observations, robot states, action logs, task or goal descriptions, navigation-state labels, and perturbation or failure metadata. This structure enables all compared approaches to access the same underlying navigation episodes while still allowing method-specific inputs. SLAM will primarily use visual/depth streams and pose ground truth. VLA components will use image observations, instructions, and action traces. RL will use the same task definition in interactive simulation for training and will be evaluated on held-out navigation variants. World Models will use sequential observations and state/action information to learn predictive navigation dynamics. This design avoids an unfair comparison based on different data distributions.

Existing datasets such as Open X-Embodiment, DROID, RL Bench, CALVIN, and RoboCasa will be treated as **secondary dataset candidates**. They may support pretraining, fine-tuning, task design, or literature-based comparison, but they will not replace the project-specific common dataset because they do not jointly satisfy the requirements of spatial tracking, semantic instruction grounding, trainable interaction, and perturbation or failure evaluation in one navigation benchmark (Open X-Embodiment Collaboration et al. 2024; Khazatsky et al. 2024; James et al. 2020; Mees et al. 2022; Soroush Nasiriany, Maddukuri, et al. 2024).

4.8 Technical Execution Environment: Isaac Lab, MuJoCo, RTX 5090, and A100

The demonstrator will use simulation software according to the strengths and hardware requirements of each tool. NVIDIA Isaac Sim and Isaac Lab are planned for high-fidelity robotic scene construction, synthetic sensor generation, domain randomization, and robot-learning workflows. Isaac Lab is especially relevant because it is designed as an open-source, GPU-accelerated modular framework for robot learning and supports reinforcement learning and imitation learning workflows (Isaac Lab Project Developers 2026; Mittal et al. 2025). Isaac Sim is suitable for robotics

simulation, testing, and synthetic data generation in physically based virtual environments (NVIDIA Corporation 2026a).

The planned hardware usage must distinguish between rendering-oriented simulation and neural model training. The NVIDIA RTX 5090 is planned as the primary local GPU for Isaac Sim / Isaac Lab rendering, scene simulation, sensor generation, synthetic-data capture, and feasible model-training experiments. By contrast, the NVIDIA A100 remains relevant for computationally intensive neural model training because the 80 GB version provides large HBM2e memory and high memory bandwidth. However, it is not suitable as the primary Isaac Sim rendering GPU because current Isaac Sim requirements state that GPUs without RT cores, including A100 and H100, are not supported (NVIDIA Corporation 2026c; NVIDIA Corporation 2026b). Consequently, the intended setup is: RTX 5090 for local Isaac Sim / Isaac Lab rendering and benchmark execution; A100 for larger training workloads, especially World Models, VLA fine-tuning where feasible, and large-batch policy-learning experiments.

MuJoCo will be considered as a complementary simulation environment. It is an open-source physics engine designed for robotics, biomechanics, graphics, animation, and optimization-oriented simulation (Google DeepMind 2026c). Standard MuJoCo is lightweight and useful for fast control experiments, reproducible RL baselines, and simplified navigation scenarios. MuJoCo can also use hardware-accelerated headless rendering through EGL on Linux (Google DeepMind 2026a). In addition, MuJoCo XLA (MJX) enables accelerator-oriented batched simulation through JAX and can run on NVIDIA GPUs, which makes it relevant for high-throughput RL experiments on an A100 (Google DeepMind 2026b). Therefore, the thesis may use Isaac Lab for the main photorealistic/sensor-rich demonstrator and MuJoCo or MJX for a simplified control-focused benchmark if this improves feasibility and reproducibility.

This hardware and software plan is part of the research design rather than an implementation detail. It makes the comparison more transparent because training time, simulation throughput, rendering feasibility, GPU memory usage, and software constraints will be documented as evaluation-relevant factors.

4.9 Evaluation Design

The evaluation is structured around shared dimensions rather than one single metric. This is necessary because the four approach classes have different primary outputs.

Evaluation dimension	Example metrics	Relevant approaches
Spatial accuracy	Absolute trajectory error, relative pose error, localization drift, map consistency	Especially SLAM, also WM and RL where spatial state is used
Navigation success	Success rate, completion time, path efficiency, number of failed attempts, recovery rate	Especially VLA and RL, also WM if used for planning

Evaluation dimension	Example metrics	Relevant approaches
Semantic correctness	Goal fulfillment, landmark or target interpretation, instruction mismatch detection	Especially VLA, also WM and rule-based goal logic
Deviation recognition and temporal prediction	Precision, recall, F1-score, AUROC, false-positive rate, detection delay after perturbation onset	Especially WM, also SLAM/RL/VLA-derived monitors
Robustness	Performance under changed lighting, obstacle positions, distractors, camera movement, layout variation, or unseen instructions	All approaches
Training, data, and compute efficiency	Required training samples, simulation steps, GPU hours, convergence behavior, inference latency, memory usage	Especially RL, VLA, and WM; SLAM mainly for runtime and parameter sensitivity
Interpretability and implementation effort	Qualitative assessment, debugging effort, required prior knowledge, explainability of errors	All approaches

Table 2: Planned evaluation dimensions and example metrics.

For deviation recognition and predictive evaluation, the most important quantitative metrics are precision, recall, F1-score, false-positive rate on nominal episodes, AUROC, and detection delay. For navigation execution, success rate, path efficiency, completion time, collision behavior, and recovery behavior are central. For spatial approaches, pose, map, and trajectory accuracy are central. For VLA approaches, goal fulfillment and semantic generalization are particularly important. For trainable approaches, the evaluation will additionally consider training stability, amount of training data, number of simulation episodes or sequences, GPU time, and reproducibility of the training results.

4.10 Fairness of the Comparison

A central methodological challenge is that SLAM, VLA, RL, and WM are not identical kinds of systems. The thesis will therefore use a capability-based comparison instead of pretending that all approaches produce the same output. The comparison will be fair by:

- using the same simulated navigation scenario and perturbation set;
- reporting method-specific metrics as well as shared high-level criteria;

- separating perception, localization, control, semantic interpretation, and predictive capabilities;
- documenting implementation assumptions, training assumptions, and model limitations;
- avoiding claims of universal superiority when a method only performs well on its natural subtask.

This design makes the thesis more scientifically defensible. It also matches robotic system design in practice: robust navigation usually depends on a pipeline or hybrid architecture, not a single isolated method.

4.11 Expected Limitations

A central limitation is the still undefined concrete use case. This is intentional at the exposé stage because the thesis is framed around simulation-based benchmarking of navigation approaches rather than one predetermined application. The final scenario must therefore be selected carefully and justified according to feasibility, benchmark value, and available simulation assets.

The second limitation is the simulation-to-real gap. Results obtained in simulation may not transfer directly to real camera data, physical robots, or operational environments. This limitation will be addressed by discussing domain randomization, sensor realism, and possible future real-world validation.

The third limitation is implementation and training feasibility. Full-scale VLA models and large World Models may exceed the scope of a master’s thesis if trained from scratch. Therefore, the thesis may use simplified, adapted, fine-tuned, or modular versions of the paradigms as long as the operationalization is documented transparently. The thesis will clearly distinguish between components trained from scratch, components fine-tuned or adapted within the thesis, and external pretrained components.

The fourth limitation is comparison fairness. Since the four paradigms have different goals and outputs, evaluation must be capability-based. The thesis should not reduce the comparison to one metric.

The fifth limitation is scenario dependence. Even if the use case is selected carefully, the results will depend on the chosen map layout, robot embodiment, sensors, action space, and perturbation types. The thesis will therefore avoid universal claims and interpret findings within the defined benchmark scope.

The sixth limitation is interpretability. Rule-based, SLAM-based, and statistical components are often easier to explain than large learned models or learned policies. This trade-off will be part of the qualitative evaluation.

4.12 Expected Outcome

The expected outcome is a working simulation-based benchmark demonstrator and a structured comparison of the selected approach classes. The result will not be a universal ranking of SLAM, VLA, RL, and WM. Instead, the thesis will provide an evidence-based matrix that links approaches to navigation capabilities, limitations, implementation effort, data demand, compute demand, and robustness under controlled simulation conditions.

5 Preliminary Outline

The master’s thesis is planned with the following preliminary structure:

1. Introduction

- (a) Motivation and problem statement
- (b) Robotic navigation relevance
- (c) Research gap
- (d) Research questions
- (e) Structure of the thesis

2. Theoretical Background

- (a) Robot navigation and simulation-based benchmarking
- (b) Deviation detection and temporal navigation monitoring
- (c) SLAM and spatial state estimation
- (d) Vision-Language-Action models
- (e) Reinforcement Learning for robot control
- (f) World Models and predictive dynamics
- (g) Simulation-based robotics research

3. Related Work

- (a) SLAM in robotic perception and monitoring
- (b) VLA and generalist robot policies
- (c) RL in simulated robot learning
- (d) World Models for prediction, planning, and deviation recognition
- (e) Existing datasets, benchmarks, and research gap

4. Research Design and Methodology

- (a) Design Science Research approach
- (b) Artifact definition
- (c) Requirements for the demonstrator
- (d) Navigation use case and assumptions
- (e) Common dataset strategy
- (f) Hardware and software execution environment
- (g) Model training scope and reproducibility
- (h) Evaluation design and fairness of comparison
- (i) Metrics and qualitative criteria

5. Design and Implementation of the Demonstrator

- (a) Simulation environment: Isaac Lab, Isaac Sim, MuJoCo, and MJX where feasible
- (b) Robot navigation scenario
- (c) Sensor and data-logging configuration
- (d) Nominal episode generation
- (e) Perturbation and failure generation
- (f) Dataset structure and preprocessing
- (g) Training pipeline, checkpoints, and experiment tracking

6. Implementation of Compared Approaches

- (a) SLAM or pose-based spatial monitoring
- (b) VLA or VLA-inspired semantic policy module, including feasible fine-tuning or adaptation
- (c) RL-based policy module, including training protocol and reward design
- (d) World Model-based predictive dynamics module, including training objective and prediction setup
- (e) Optional hybrid architecture

7. Evaluation

- (a) Experimental setup
- (b) Spatial accuracy results
- (c) Navigation success and robustness results
- (d) Semantic correctness results
- (e) Deviation recognition and temporal prediction results
- (f) Detection delay and early-warning capability
- (g) Runtime, training effort, data efficiency, and implementation effort
- (h) Failure-case analysis

8. Discussion

- (a) Interpretation of results
- (b) Comparison with hypotheses
- (c) Implications for robot navigation systems
- (d) Hybrid architecture potential
- (e) Limitations and sim-to-real considerations

9. Conclusion and Outlook

- (a) Summary of contributions
- (b) Answering the research questions
- (c) Future work

6 Time Plan

The following time plan assumes a six-month master’s thesis period. It is presented on a landscape page to show the complete project cycle from technical familiarization to final submission. The plan deliberately does not start with research-question development, because the research questions and general scope are assumed to be fixed at the beginning. Instead, the early project phase focuses on becoming productive in the technical working environment, building the simulation setup for the Unitree Go2 robot dog, and reviewing the relevant literature in parallel. The implementation phase is planned as a longer iterative cycle: the system must first become runnable, then initial simulations and early comparisons can begin, and only afterward can systematic benchmarking be refined and executed.

Table 3: Extended landscape timeline for the complete master’s thesis project cycle.

Phase	Period	Project step	Main focus	Activities, milestones, and outputs
1	Week 1–2	Technical familiarization	Isaac Sim / Isaac Lab working environment	Become familiar with Isaac Sim, Isaac Lab, the simulation workflow, the local RTX 5090 setup, repository structure, experiment tracking, and available Unitree Go2 assets or robot-description files. Output: working development environment, documented setup steps, first runnable tests, and feasibility notes.
2	Week 1–4	Literature grounding	Focused theoretical and technical review	Review Design Science Research, robot-navigation benchmarking, quadruped navigation, SLAM, VLA/VLA-inspired robot policies, RL, World Models, Isaac Lab, MuJoCo/MJX, sim-to-real constraints, and existing robot-learning datasets. Output: literature map, annotated bibliography, related-work structure, and refined evaluation dimensions.
3	Week 3–5	Simulation environment construction	Training environment for the Unitree Go2	Build the initial simulation environment in which the robot dog can be trained and evaluated. Define layout, spawn and goal conditions, basic sensors, obstacles, reset logic, and episode boundaries while keeping the exact use case flexible. Output: first navigation environment draft with reproducible scene configuration.
4	Week 5–7	First runnable implementation	End-to-end benchmark skeleton	Connect simulation, control interface, logging, episode execution, and initial baseline components into a runnable pipeline. The goal is not final performance yet, but a stable system that can execute complete episodes and record the required data. Output: runnable prototype, logging schema, pilot episodes, and bug list.
5	Week 8–10	Initial simulation and early comparison	Pilot experiments and benchmark refinement	Start simulation experiments once the system is runnable. Compare early baseline behavior, simple navigation policies, and first approach-specific components to identify missing metrics, unstable assumptions, and bottlenecks. Output: pilot result tables, first comparison notes, refined metrics, and updated implementation priorities.
6	Week 11–14	Iterative implementation and training I	Main development cycle	Extend and improve the implementation iteratively. Train or adapt the RL navigation policy, configure SLAM or pose-based baselines, develop feasible World Model components, and test VLA/VLA-inspired modules where possible. Output: model versions, checkpoints, training logs, and documented hyperparameters.
7	Week 15–18	Iterative implementation and training II	Repeated simulation, debugging, and comparison	Continue the implementation cycle by alternating between simulation runs, debugging, metric revision, perturbation design, retraining, and comparison. Output: stable simulation setup, improved approach implementations, preliminary comparative findings, and finalized benchmark protocol.
8	Week 19–20	Systematic benchmarking	Full benchmark execution	Run the structured benchmark across selected approaches, scenario variants, perturbations, and evaluation settings. Evaluate success rate, trajectory quality, robustness, localization consistency, goal-conditioned behavior where applicable, compute effort, data demand, implementation effort, and reproducibility. Output: complete result tables, plots, failure-case collection, and comparative evaluation matrix.
9	Week 21–22	Evaluation and interpretation	Analysis and synthesis	Interpret the benchmark results in relation to RQ1, RQ2, and the reduced hypothesis set. Identify which capabilities are supported by each approach, where limitations appear, and whether hybrid or capability-based combinations are more plausible than isolated methods. Output: results chapter draft, discussion outline, limitation analysis, and refined contribution statement.
10	Week 23–24	Writing and finalization	Thesis completion and submission	Complete and revise the introduction, related work, methodology, implementation, results, discussion, conclusion, figures, tables, citations, reproducibility notes, and artifact documentation. Output: final thesis, final artifact documentation, proofreading checklist, and submitted master’s thesis.

7 Expected Contribution

The thesis is expected to make a scientific, practical, and methodological contribution.

From a scientific perspective, it provides a structured comparison of SLAM, VLA, RL, and World Model approaches in one simulation-based robot navigation scenario. Instead of evaluating each paradigm in isolation, the thesis investigates their relative suitability under shared simulation conditions and differentiated evaluation criteria.

From a practical perspective, the thesis delivers a simulation-based demonstrator that can generate, perturb, label, train models on, and evaluate robot navigation episodes using a common episode dataset. The demonstrator can serve as a foundation for later real-world validation, edge deployment, or navigation benchmarking and robotic evaluation prototypes. In addition, it documents the practical effort required to train or adapt each learning-based approach, which is important for practical feasibility assessments.

From a methodological perspective, the thesis demonstrates how Design Science Research can be used to build and evaluate an artifact for emerging robotics and AI technologies. The work contributes a capability-based benchmark logic that avoids unfair one-dimensional comparisons and instead clarifies where each approach provides value.

8 Preliminary Literature

The preliminary literature base combines several groups of sources. Design Science Research provides the methodological foundation for artifact-oriented research (Hevner et al. 2004; Peffers et al. 2007). Deviation detection, benchmarking, and statistical process control provide the background for monitoring and evaluation (Bergmann et al. 2019; Montgomery 2019). SLAM literature provides the spatial perception and localization foundation (Durrant-Whyte and Bailey 2006; Campos et al. 2021). Reinforcement Learning literature provides the control-learning foundation (Sutton and Barto 2018; Mnih et al. 2015; Schulman et al. 2017). VLA and generalist robot-policy research provide the semantic robot-navigation and control foundation (Zitkovich et al. 2023; Open X-Embodiment Collaboration et al. 2024; Kim et al. 2025; Octo Model Team et al. 2024). World Model and JEPA-style research provide the predictive dynamics and planning foundation (Ha and Schmidhuber 2018; Hafner et al. 2023; Bardes et al. 2024; Assran, Bardes, et al. 2025). Existing robot-learning datasets and benchmarks provide the background for the common dataset strategy (Open X-Embodiment Collaboration et al. 2024; Khazatsky et al. 2024; James et al. 2020; Mees et al. 2022; Soroush Nasiriany, Maddukuri, et al. 2024; Soroush Nasiriany, Sepehr Nasiriany, et al. 2026). NVIDIA Isaac Sim, Isaac Lab, MuJoCo, and MJX provide the planned simulation and robot-learning context, while RTX 5090/A100 hardware constraints define the feasible execution environment (NVIDIA Corporation 2026a; Isaac Lab Project Developers 2026; NVIDIA Corporation 2026b; Google DeepMind 2026c; Google DeepMind 2026b; NVIDIA Corporation 2026c).

The final thesis will expand this literature base with more specific sources on temporal deviation detection, robotic foundation models, domain randomization, sim-to-real transfer, uncertainty estimation, and robot navigation benchmarking.

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